main.c

Run ▶

```c
#include  <stdio.h>

int main()
{
    int at[10], bt[10], pr[10];
    int n, i, j, temp, time = 0, count, over = 0, sum_wait = 0, sum_turnaround = 0, start;
    float avgwait, avgturn;

    printf("Enter the number of processes\n");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("Enter the arrival time and execution time for process %d\n", i + 1);
        scanf("%d%d", &at[i], &bt[i]);
        pr[i] = i + 1;
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (at[i] > at[j])
            {
                temp = at[i]; at[i] = at[j]; at[j] = temp;
                temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
                temp = pr[i]; pr[i] = pr[j]; pr[j] = temp;
            }
        }
    }

    printf("\n\nProcess\t|Arrival time\t|Execution time\t|Start time\t|End time\t|waiting time\t|Turnaround time\n\n");

    while (over < n)
    {
        count = 0;
        for (i = over; i < n; i++)
        {
```

Output

```
Enter the number of processes
3
Enter the arrival time and execution time for process 1
0 5
Enter the arrival time and execution time for process 2
1 3
Enter the arrival time and execution time for process 3
2 8


Process |Arrival time   |Execution time |Start time     |
End time|waiting time   |Turnaround time

p[1]    |       0       |       5       |       0       |
5       |       0       |       5
p[2]    |       1       |       3       |       5       |
8       |       4       |       7
p[3]    |       2       |       8       |       8       |
16      |       6       |       14
Average waiting time is 3.333333
Average turnaround time is 8.666667


=== Code Execution Successful ===
```

```c
            if (at[i] <= time)
                count++;
            else
                break;
        }

        if (count > 1)
        {
            for (i = over; i < over + count - 1; i++)
            {
                for (j = i + 1; j < over + count; j++)
                {
                    if (bt[i] > bt[j])
                    {
                        temp = at[i]; at[i] = at[j]; at[j] = temp;
                        temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
                        temp = pr[i]; pr[i] = pr[j]; pr[j] = temp;
                    }
                }
            }
        }

        start = time;
        time += bt[over];

        printf("p[%d]\t|\t%d\t|\t%d\t|\t%d\t|\t%d\t|\t%d\t|\t%d\n",
               pr[over], at[over], bt[over], start, time,
               time - at[over] - bt[over], time - at[over]);

        sum_wait += time - at[over] - bt[over];
        sum_turnaround += time - at[over];
        over++;
    }

    avgwait = (float)sum_wait / (float)n;
    avgturn = (float)sum_turnaround / (float)n;

    printf("Average waiting time is %f\n", avgwait);
    printf("Average turnaround time is %f\n", avgturn);

    return 0;
}
```